

PRACTICE SHEET

1. The angles of elevation of the top of a tower from the top and foot of a pole are respectively 30° and 45° is h_T is the height of the tower and h_p is the height of the pole, then which of the following are correct?
 1. $\frac{2h_p h_T}{3 + \sqrt{3}} = h_p^2$
 2. $\frac{h_T - h_p}{\sqrt{3} + 1} = \frac{h_p}{2}$
 3. $\frac{2(h_p + h_T)}{h_p} = 4 + \sqrt{3}$

Select the correct answer using the code given below:

(a) 1 and 3 only (b) 2 and 3 only
(c) 1 and 2 only (d) 1, 2 and 3
2. From the top of a lighthouse, 100 m high, the angle of depression of a boat is $\tan^{-1}\left(\frac{5}{12}\right)$. What is the distance between the boat and the lighthouse?

(a) 120 m (b) 180 m
(c) 240 m (d) 360 m
3. The top of a hill when observed from the top and bottom of a building of height h is at angles of elevation p and q respectively. What is the height of the hill?

(a) $\frac{h \cot q}{\cot q - \cot p}$ (b) $\frac{h \cot p}{\cot p - \cot q}$
(c) $\frac{2h \tan p}{\tan p - \tan q}$ (d) $\frac{2h \tan q}{\tan q - \tan p}$
4. A moving boat is observed from the top of a cliff of 150 m height. The angle of depression of the boat changes from 60° to 45° in 2 minutes. What is the speed of the boat in metres per hour?

(a) $\frac{4500}{\sqrt{3}}$ (b) $\frac{4500(\sqrt{3}-1)}{\sqrt{3}}$
(c) $4500\sqrt{3}$ (d) $\frac{4500(\sqrt{3}+1)}{\sqrt{3}}$
5. Two poles are 10 m and 20 m high. The line joining their tops makes an angle of 15° with the horizontal. The distance between the poles is approximately equal to

(a) 36.3 m (b) 37.3 m
(c) 38.3 m (d) 39.3 m
6. A vertical tower standing on a leveled field is mounted with a vertical flag staff of length 3 m. From a point on the field, the angles of elevation of the bottom and tip of the flag staff are 30° and 45° , respectively. Which one of the following gives the best approximation to the height of the tower?

(a) 3.90 m (b) 4.00 m
(c) 4.10 m (d) 4.25 m
7. The angle of elevation of the top of a tower from a point 20 m away from its base is 45° . What is the height of the tower?

(a) 10 m (b) 20 m
(c) 30 m (d) 40 m
8. The angles of elevation of the top of a tower standing on a horizontal plane from two points on a line passing through the foot of the tower at distances 49 m and 36 m are 43° and 47° respectively. What is the height of the tower?

(a) 40 m (b) 42 m
(c) 45 m (d) 47 m
9. A lamp post stands on a horizontal plane. From a point situated at a distance 150 m from its foot, the angle of elevation of the top is 30° . What is the height of the lamp post?

(a) 50 m (b) $50\sqrt{3}$ m
(c) $\frac{50}{\sqrt{3}}$ m (d) 100 m
10. From an aeroplane above a straight road the angles of depression of two positions at a distance 20 m apart on the road are observed to be 30° and 45° . The height of the aeroplane above the ground is

(a) $10\sqrt{3}$ m (b) $10(\sqrt{3}-1)$ m
(c) $10(\sqrt{3}+1)$ m (d) 20 m
11. If the angles of elevation of the top of a tower from two places situated at distances 21 m and x m from the base of the tower are 45° and 60° respectively, then what is the value of x ?

- (a) $7\sqrt{3}m$ (b) $7 - \sqrt{3}m$
(c) $7 + \sqrt{3}m$ (d) 14
- 12.** A person standing on the bank of a river observes that the angle subtended by a tree on the opposite of bank is 60° . When he returns 40 m from the bank, he finds the angle to be 30° . What is the breadth of the river?
(a) 60 m (b) 40 m
(c) 30 m (d) 20 m
- 13.** The angle of elevation of the top of a tower of height H from the foot of another tower in the same plane is 60° and the angle of elevation of the top of the second tower from the foot of the first tower is 30° . If is the height of the other tower, then which one of the following is correct?
(a) $H=2h$ (b) $H = \sqrt{3} h$
(c) $H = 3h$ (d) None of these
- 14.** A man walks 10 m towards a lamp post and notices that the angle of elevation of the top of the post increases from 30° to 45° . the height of the lamp post is
(a) 10 m (b) $(5\sqrt{3}+5)m$
(c) $(5\sqrt{3}-5)m$ (d) $(10\sqrt{3}+10)m$
- 15.** The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30° than when it is 60° . The height of the tower is
(a) 25 m (b) $25\sqrt{3} m$
(c) 50 m (d) none of these
- 16.** The top of a hill observed from the top and bottom of a building of height h is at angles of elevation α and β respectively. The height of the hill is
(a) $\frac{h \cot \beta}{\cot \beta - \cot \alpha}$ (b) $\frac{h \cot \alpha}{\cot \alpha - \cot \beta}$
(c) $\frac{h \cot \alpha}{\tan \alpha - \tan \beta}$ (d) None of these
- 17.** From the top of a lighthouse 70 high with its base at sea level, the angle of depression of a boat is 15° . The distance of the boat from the foot of the lighthouse is
(a) $70(2 - \sqrt{3})m$ (b) $70(2 + \sqrt{3})m$
(c) $70(3 - \sqrt{3})m$ (d) $70(3 + \sqrt{3})m$
- 18.** Two poles are 10 m and 20 m high. The line joining their tips makes an angle of 15° with the horizontal. What is the distance between the poles?
(a) $10(\sqrt{3}-1)m$ (b) $5(4+2\sqrt{3})m$
(c) $20(\sqrt{3}+1)$ (d) $10(\sqrt{3}+1)m$
- 19.** The angle of elevation of a tower at a level ground is 30° . The angle of elevation becomes θ when moved 10 m towards the tower. If the height of tower is $5\sqrt{3} m$, then what is the value of θ ?
(a) 45° (b) 60°
(c) 75° (d) None
- 20.** From the top of a building of height h m, the angle of depression of an object on the ground is θ . What is the distance (in m) of the object from the foot of the building?
(a) $h \cot \theta$ (b) $h \tan \theta$
(c) $h \cos \theta$ (d) $h \sin \theta$
- 21.** The angle of elevation of the tip of a flag staff from a point 10 m due South of its base is 60° . What is the height of the flag staff correct to nearest meter?
(a) 15 m (b) 16 m
(c) 17 m (d) 18 m
- 22.** What is the angle subtended by 1 m pole at distance 1 km on the ground in sexagesimal measure?
(a) $\frac{9}{50\pi}$ degree (b) $\frac{9}{5\pi}$ degree
(c) 3.4 min (d) 3.5 min
- 23.** A tower of height 15 m stands vertically on the ground. From a point on the ground the angle of elevation of the top of the tower is found to be 30° . What is the distance of the point from the foot of the tower?
(a) $15\sqrt{3} m$ (b) $10\sqrt{3}$
(c) $5\sqrt{3} m$ (d) 30
- 24.** At a point 15 m away from the base of a 15 m high house, the angle of elevation of the top is
(a) 90° (b) 60°
(c) 45° (d) 30°
- 25.** A vertical tower stands on a horizontal plane and is surmounted by a vertical flag staff of elevation of the bottom of the flag

staff is β and that of the top is α . What is the height of the tower?

(a) $\frac{h \tan \beta}{\tan \alpha}$ (b) $\frac{h \tan \beta}{\tan \alpha + \tan \beta}$

(c) $\frac{h \cos \beta}{\tan \alpha - \tan \beta}$ (d) $\frac{h}{\cos(\alpha - \beta)}$

- 26.** An aeroplane flying at a height of 300 m above the ground passes vertically above another plane at an instant when the angles of elevation of two planes from the same point on the ground are 60° and 45° , respectively. What is the height of the lower plane from the ground?

(a) 50 m (b) $\frac{100}{\sqrt{3}}$ m
(c) $100\sqrt{3}$ m (d) $150(\sqrt{3}+1)$ m

- 27.** A man standing on the bank of a river observes that the angle of elevation of the top of a tree just on the opposite bank is 60° . The angle of elevation is 30° from a point at a distance y m from the bank. What is the height of the tree?

(a) y m (b) $2y$ m
(c) $\frac{\sqrt{3}y}{2}$ m (d) $\frac{y}{2}$ m

- 28.** Two poles are 10 m and 20 m high. The line joining their tops makes an angle of 15° with the horizontal. What is the approximate distance between the poles?

(a) 35.3 m (b) 37.3 m
(c) 41 m (d) 44 m

- 29.** From the top of a lighthouse 120 m above the sea, the angle of depression of a boat is 15° . What is the distance of the boat from the lighthouse?

(a) 400 m (b) 421 m
(c) 444 m (d) 460 m

- 30.** The angle of elevation of the top of a flag post from a point 5 m away from its base is 75° . What is the approximate height of the flag post?

(a) 15 m (b) 17 m
(c) 19 m (d) 21 m

ANSWER KEY

1.	(c)	2.	(c)	3.	(b)	4.	(b)	5.	(b)	6.	(c)	7.	(b)	8.	(b)	9.	(b)	10.	(c)
11.	(c)	12.	(d)	13.	(c)	14.	(b)	15.	(b)	16.	(b)	17.	(b)	18.	(b)	19.	(b)	20.	(a)
21.	(c)	22.	(a)	23.	(a)	24.	(c)	25.	(a)	26.	(c)	27.	(c)	28.	(b)	29.	(c)	30.	(c)

CDS PYQ

DIRECTIONS (for next four): Read the following information carefully to answer the questions that follow As seen from the top and bottom of a building of height h m, the angles of elevation of the top of a tower of height $\frac{3+\sqrt{3}}{2}h$ are α and β , respectively.

[CDS 2013 (I)]

1. If $\beta = 30^\circ$, then what is the value of $\tan \alpha$?
(a) $1/2$ (b) $1/3$
(c) $1/4$ (d) none of these
2. If $\alpha = 30^\circ$, then what is the value of $\tan \beta$?
(a) 1 (b) $1/2$
(c) $1/3$ (d) none of these
3. If $\alpha = 30^\circ$ and $h = 30$ m, then what is the distance between the base of the building and the base of the tower?
(a) $15+15\sqrt{3}$ m (b) $30+15\sqrt{3}$ m
(c) $45+15\sqrt{3}$ m (d) None of these
4. If $\beta = 30^\circ$ and if θ is the angle of depression of the foot of the tower as seen from the top of the building, then what is the value of $\tan \theta$?
(a) $\frac{(3-\sqrt{3})}{3\sqrt{3}}$ (b) $\frac{(3+\sqrt{3})}{3\sqrt{3}}$
(c) $\frac{(2-\sqrt{3})}{3\sqrt{3}}$ (d) none of these
5. A spherical balloon of radius r subtends angle 60° at the eye of an observer. If the angle of elevation of its centre is 60° and h is the height of the centre of the balloon, then which one of the following is correct?

[CDS 2013 (II)]

- | | |
|------------|-------------------|
| (a) $h=r$ | (b) $h=\sqrt{2}r$ |
| (c) $h=3r$ | (d) $h=2r$ |

6. What is the angle of elevation of the Sun, when the shadow of a pole of height x m is $\frac{x}{\sqrt{3}}$ m?

[CDS 2013 (II)]

- | | |
|----------------|----------------|
| (a) 30° | (b) 45° |
| (c) 60° | (d) 75° |

7. The shadow of a tower standing on a level plane is found to be 50 m longer when the Sun's elevation is 30° , then when it is 60° . What is the height of the tower?

[CDS 2014 (I)]

- | | |
|---------------------|--------------------|
| (a) 25m | (b) $25\sqrt{3}$ m |
| (c) $25/\sqrt{3}$ m | (d) 30 m |

8. The angle of elevation of the top of a tower 30 m high from the foot of another tower in the same plane is 60° and the angle of elevation of the top of the second tower from the foot of the first tower is 30° . The distance between the two towers in m times the height of the shorter tower. What is m equal to?

[CDS 2014 (I)]

- | | |
|------------------|------------------|
| (a) $\sqrt{2}$ m | (b) $\sqrt{3}$ m |
| (c) $1/2$ | (d) $1/3$ |

9. From a certain point on a straight road, a person observe a tower in the West direction at a distance of 200 m. He walks some distance along the road and finds that the same tower is 300 m South of him. What is the shortest distance of the tower from the road?

[CDS 2014 (II)]

- | | |
|-------------------------------|-------------------------------|
| (a) $\frac{300}{\sqrt{13}}$ m | (b) $\frac{500}{\sqrt{13}}$ m |
| (c) $\frac{600}{\sqrt{13}}$ m | (d) $\frac{900}{\sqrt{13}}$ m |

10. If from the top of a post a string twice the length of the post is stretched tight to a point on the ground, then what angle will the string make with the post?

[CDS 2014 (II)]

- | | |
|---------------|-------------|
| (a) $\pi/6$ | (b) $\pi/4$ |
| (c) $5\pi/12$ | (d) $\pi/3$ |

11. The angles of elevation of the top of a tower from two points P and Q at distances m^2 and n^2 respectively, from the base and in the same straight line with it are complementary. The height of the tower is

[CDS 2015 (I)]

- | | |
|------------------|----------------|
| (a) $(mn)^{1/2}$ | (b) $mn^{1/2}$ |
| (c) $m^{1/2}n$ | (d) mn |

- 12.** The angle of elevation of a cloud from a point 200 m above a lake is 30° and the angle of depression of its reflection in the lake is 60° . The height of the cloud is
[CDS 2015 (I)]
(a) 200 m (b) 300 m
(c) 400 m (d) 600 m
- 13.** From the top of a tower, the angles of depression of two objects P and Q (situated on the ground on the same side of the tower) separated at a distance of $100(3-\sqrt{3})$ m are 45° and 60° respectively. The height of the tower is
[CDS 2015 (I)]
(a) 200 m (b) 250 m
(c) 300 m (d) none of these
- 14.** An aeroplane flying at a height of 3000 m passes vertically above another aeroplane at an instant when the angles of elevation of the two planes from some point on the ground are 60° and 45° respectively. Then the vertical distance between the two planes is
[CDS 2015 (II)]
(a) $1000(\sqrt{3}-1)$ m (b) $1000\sqrt{3}$ m
(c) $(1000(3-\sqrt{3}))$ m (d) $3000\sqrt{3}$ m
- 15.** A pole is standing erect on the ground which is horizontal. The tip of the pole is tied tight with a rope of length $\sqrt{12}$ m to a point on the ground. If the rope is making 30° with the horizontal, then the height of the pole is
[CDS 2015 (II)]
(a) $2\sqrt{3}$ m (b) $3\sqrt{2}$ m
(c) 3 m (d) $\sqrt{3}$ m
- 16.** Two observers are stationed due north of a tower (of height x metre) at a distance y metre from each other. The angles of elevation of the tower observed by them are 30° and 45° respectively. Then x/y is equal to
[CDS 2016 (I)]
(a) $\frac{\sqrt{2}-1}{2}$ (b) $\frac{\sqrt{2}-1}{2}$
(c) $\frac{\sqrt{3}+1}{2}$ (d) 1
- 17.** A man from the top of a 100 m high tower sees a car moving towards the tower at an angle of depression 30° . After some time, the angle of depression becomes 60° . What is the distance travelled by the car during this time?
[CDS 2016 (II)]
(a) $100\sqrt{3}$ m (b) $\frac{200\sqrt{3}}{3}$ m
(c) $\frac{100\sqrt{3}}{3}$ m (d) $200\sqrt{3}$ m
- 18.** Two men on either side of a tower 75 m high observe the angle of elevation of the top of the tower to be 30° and 60° . What is the distance between the two men?
[CDS 2016 (II)]
(a) $100\sqrt{3}$ m (b) $75\sqrt{3}$ m
(c) $\frac{100\sqrt{3}}{3}$ m (d) $60\sqrt{3}$ m
- 19.** If the length of the shadow of a tower is equal to its height, then what is the Sun's altitude at that time?
[CDS 2016 (II)]
(a) 15° (b) 30°
(c) 45° (d) 60°
- 20.** A pole stands vertically inside a triangular park ABC. If the angle of elevation of the top of the pole from each corner of the park is same, then in the triangle ABC, the foot of the pole is at the
[CDS 2016 (II)]
(a) Centroid (b) Circumference
(c) Incentre (d) Orthocentre
- 21.** An aeroplane flying at a height of 300 m above the ground passes vertically above another plane at an instant when the angles of elevation of the two planes from the same point on the ground are 60° and 45° respectively. What is the height of the lower plane from the ground?
[CDS 2017 (I)]
(a) $100\sqrt{3}$ m (b) $\frac{100}{\sqrt{3}}$ m
(c) $50\sqrt{3}$ m (d) $150(\sqrt{3}+1)$ m
- 22.** From the top of a building 90 m high, the angles of depression of the top and the bottom of a tree are 30° and 45° respectively. What is the height of the tree?
[CDS 2017 (I)]
(a) $30\sqrt{3}$ m (b) $90-30\sqrt{3}$ m
(c) $90+30\sqrt{3}$ m (d) $60+30\sqrt{3}$ m
- 23.** From an aeroplane vertically over a straight horizontal road, the angles of depression of

two consecutive kilometre- stones on the opposite sides of the aeroplane are observed to be A and B. The height of the aeroplane above the road is

[CDS 2017 (I)]

- | | |
|---|---|
| (a) $\frac{\tan \alpha + \tan \beta}{\tan \alpha \tan \beta}$ | (b) $\frac{\tan \alpha \tan \beta}{\tan \alpha + \tan \beta}$ |
| (c) $\frac{\cot \alpha \cot \beta}{\cot \alpha + \cot \beta}$ | (d) $\frac{\cot \alpha + \cot \beta}{\cot \alpha \cot \beta}$ |

24. Let AB represent a building of height h metre with A being its top, B being its bottom. Let A'B' represent a tower of height (h + x) metre (x > 0) with A' being its top and B' being its bottom. Let BB' = d metre. Let the angle of elevation of A' as seen from A be 45°.

Consider the following statements:

Statement I: h + x > d

Statement II: The angle of depression of B as seen from A' is less than 45° Which one of the following is correct in respect of the above statements?

[CDS 2017 (II)]

- (a) Both Statement I and Statement II are true and Statement II is the correct explanation of Statement I
 (b) Both Statement I and Statement II are true but Statement II is not the correct explanation of Statement I
 (c) Statement I is true but Statement II is false
 (d) Statement I is false but Statement II is true 58.

25. A man, standing at a point X on the bank XY of a river that cannot be crossed, observes a tower to be N α° E on the opposite parallel bank. He then walks 200 m along the bank to the point Y towards East, and finds the tower to be N β° W. From these observations, the breadth of the river will be (Given that $\tan \alpha^\circ = 2$ and $\tan \beta^\circ = 0.5$)

[CDS 2017 (II)]

- | | |
|----------|----------|
| (a) 60 m | (b) 70 m |
| (c) 80 m | (d) 90 m |

26. Each side of a square subtends an angle of 60° at the tip of a tower of height h metres standing at the centre of the square. If l is the length of each side of the square, then what is h² equal to?

[CDS 2018 (I)]

- | | |
|----------------------|----------------------|
| (a) $2l^2$ | (b) $\frac{l^2}{2}$ |
| (c) $\frac{3l^2}{2}$ | (d) $\frac{2l^2}{3}$ |

27. From a height of h units, a man observes the angle of elevation as α and angle of depression as β of the top and the bottom respectively of a tower of height H (> 4h). To what further height should he climb so that the values of angle of elevation and angle of depression get interchanged for the top and bottom of the tower?

[CDS 2018 (I)]

- | | |
|----------------|----------------|
| (a) H-h units | (b) H-2h units |
| (c) H-3h units | (d) H-4h units |

28. On the top of a hemispherical dome of radius r, there stands a flag of height h. From a point on the ground, the elevation of the top of the flag is 30°. After moving a distance d towards the dome, when the flag is just visible, the elevation is 45°. The ratio of h to r is equal to:

[CDS 2018 (II)]

- | | |
|-------------------------------------|---|
| (a) $\sqrt{2} - 1$ | (b) $\frac{\sqrt{3}+1}{2\sqrt{2}}$ |
| (c) $\frac{\sqrt{3}+1}{2\sqrt{2}}d$ | (d) $\frac{(\sqrt{3}+1)(\sqrt{2}-1)}{2\sqrt{2}}d$ |

29. Consider a regular hexagon ABCDEF. Two towers are situated at B and C. The angle of elevation from A to the top of the tower at B is 30°, and the angle of elevation to the top of the tower at C is 45°. What is the height of towers at B and C?

[CDS-2019-1]

- | | |
|-------------------|--------------------|
| (a) 1: $\sqrt{3}$ | (b) 1:3 |
| (c) 1:2 | (d) 1:2 $\sqrt{3}$ |

30. The angles of elevation of the tops of two pillars of heights h and 2h from a point P on the line joining the feet of the two pillars are complementary. If the distances of the foot of the pillars from the point P are x and y respectively, then which one of the following is correct?

[CDS-2019-2]

- | | |
|--------------------|----------------------|
| (a) $2h^2 = x^2 y$ | (b) $2h^2 = xy^2$ |
| (c) $2h^2 = xy$ | (d) $2h^2 = x^2 y^2$ |

31. A tree of height 15 m is broken by wind in such a way that its top touches the ground and makes an angle 30° with the ground.

What is the height from the ground to the point where tree is broken?

[CDS-2020-1]

- (a) 10m (b) 7m
(c) 5m (d) 3m

32. On a plane area there are two vertical towers separated by 100 feet apart. The shorter tower is 40 feet tall. A pole of length 6 feet stands on the line joining the base of two towers so that the tip of the towers and tip of the pole are also on the same line. If the distance of the pole from the shorter tower is 75feet, then what is the height of the taller tower (approximately)?

[CDS-2020-1]

- (a) 85feet (b) 110feet
(c) 125feet (d) 140fet

33. The angles of elevation of the top of a tower from two points at distances p and q from the base and on the same straight line are 27° and 63° respectively. What is the height of the tower?

[CDS-2020-1]

- (a) pq (b) $\sqrt{pq}$
(c) $\frac{pq}{2}$ (d) $\frac{\sqrt{pq}}{2}$

34. A pole on the ground leans at 60° with the vertical. At a point x metre away from the base of the pole on the ground, two halves of the pole subtended the same angle. If the pole and point are in the same vertical plane, then what is the length of the pole?

[CDS-2021-1]

- (a) $\sqrt{2}x$ metre (b) $\sqrt{3}x$ metre
(c) 2x metre (d) $2\sqrt{2}x$ metre

35. A vertical tower standing at the corner of a rectangular field subtends angles of 60° and 45° at the two nearer corners. If θ is the angle that the tower subtends at the farthest corner, then what is $\cot\theta$ equal to?

[CDS-2021-1]

- (a) $\frac{1}{2}$ (b) 2
(c) $\frac{2}{\sqrt{3}}$ (d) $\frac{4}{\sqrt{3}}$

36. A ladder 10m long reaches a point 10m below the top of a vertical flagstaff. From foot of the ladder, the elevation of top of the

flagstaff is 60° . What is the height of flagstaff?

[CDS-2021-2]

- (a) 12m (b) 15m
(c) 16m (d) 20m

37. From an aero-plane flying above a river at an altitude of 1200m, it is observed that the angles of depression of opposite points on the two banks of a river are 30° and θ . If the width of the river is 3000m, then which one of the following is correct?

[CDS-2021-2]

- (a) $\theta < 30^\circ$ (b) $30^\circ < \theta < 45^\circ$
(c) $45^\circ < \theta < 60^\circ$ (d) $60^\circ < \theta < 90^\circ$

38. An aeroplane is observed to the approaching the airport. It is at a distance of 10km from the point of observation on the ground and makes an angle of elevation θ . If the aeroplane is at a height of 8km above the ground, then which one of the following is correct?

[CDS-2022-1]

- (a) $0 < \theta < 30^\circ$ (b) $30^\circ < \theta < 45^\circ$
(c) $45^\circ < \theta < 60^\circ$ (d) $60^\circ < \theta < 90^\circ$

39. The angle of elevation of the top of a tower of height x meter from a point on the ground is found to be 60° . By going y meter away from that point, it becomes 30° . Which one of the following relations is correct?

[CDS-2022-1]

- (a) $x = y$ (b) $2x = 3y$
(c) $2x = \sqrt{3}y$ (d) $2y = \sqrt{3}x$

Consider the following for the next two (02) items that follow:

The angle of elevation of a cloud at C from a point (P), H metres above the surface of a lake is 30° . The height of the cloud from the surface of the lake is 2H metres. Let θ be the angle of depression of the reflection of the cloud in the lake from the point P.

40. What is the value of θ ?

[CDS 2022 (II)]

- (a) 30°
(b) 45°
(c) 60°
(d) cannot be determined due to insufficient data

41. What is PC equal to?

[CDS 2022 (II)]

- (a) H metres (b) $\sqrt{2}H$ metres
(c) $\sqrt{3}H$ metres (d) $2H$ metres

42. A person on the top of a vertical tower observes a car moving at a uniform speed coming directly towards it. If it takes 6 minutes for the angle of depression to change from 30° to 45° , and further t minutes to reach the tower, which one of the following is correct?

[CDS - 2023 (1)]

- (a) $7.7 < t < 8$ (b) $8 < t < 8.3$
(c) $8.3 < t < 8.6$ (d) $8.6 < t < 8.9$

43. A woman is standing on the deck of a ship, which is h (in metres) above water level. She observes the angle of elevation of the top of a tower as 60° and the angle of depression of the base of the tower as 30° . What is the height of the tower?

[CDS - 2023 (1)]

- (a) $2h$ (b) $3h$
(c) $4h$ (d) $5h$

Consider the following for the next two (02) items that follow:

A flagstaff stands on the top of a vertical tower. The angle of elevation of the top of the flagstaff from a certain place on the same horizontal level with the base of the tower is found to be α . Advancing a distance d towards the tower in the same horizontal plane, the angle of elevation of the top of the flagstaff is observed to be β and that of the top of the tower is observed to be γ . Let H be the height of the top of the flagstaff from the base of the tower and h be the height of the tower.

44. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $H \tan \gamma - h \tan \beta = 0$ (b) $h \tan \gamma - H \tan \beta = 0$

- (c) $H \tan \gamma - h \tan \alpha = 0$ (d) $h \tan \gamma - H \tan \alpha = 0$

45. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $d = \frac{H(\cot \alpha - \cot \beta)}{2}$ (b) $d = \frac{H(\tan \alpha - \tan \beta)}{2}$
(c) $d = H(\cot \alpha - \cot \beta)$ (d) $d = H(\tan \alpha - \tan \beta)$

Consider the following for the next two (02) items that follow:

AB is a straight road leading to the foot P of a tower of height h . Q is at distance x from P and R is at a distance y from Q. (R is farther from P and Q. R, Q are on the same side). The angle of elevation of the top of the tower at Q is twice of that at R. (use the formula $\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$)

46. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $x=y$
(b) $x < y$
(c) $x > y$
(d) cannot be concluded due to insufficient data

47. Which one of the following is correct?

[CDS-2023 (2)]

- (a) $h^2 = x^2 - y^2$ (b) $h^2 = x^2 + y^2$
(c) $h^2 = 2(y^2 - x^2)$ (d) $h^2 = y^2 - x^2$

48. The angles of elevation of the top of a tower from two points A and B at a distance of x m and $(x + 5)$ m from the base of the tower of height 6 m and in the same straight line with it are complementary. What is the value of x ?

[CDS-2024 (1)]

- (a) 4 m (b) 5 m
(c) 6 m (d) 9 m

ANSWER KEY

1.	b	2.	a	3.	c	4.	a	5.	c	6.	c	7.	b	8.	b	9.	c	10.	d
11.	d	12.	c	13.	c	14.	c	15.	d	16.	c	17.	b	18.	a	19.	c	20.	a
21.	a	22.	b	23.	b	24.	c	25.	c	26.	b	27.	c	28.	a	29.	b	30.	c
31.	c	32.	a	33.	b	34.	b	35.	c	36.	b	37.	c	38.	b	39.	c	40.	c
41.	d	42.	b	43.	c	44.	a	45.	c	46.	b	47.	d	48.	a				

